

Exercise 03

PART A

For each of the following questions there are four choices marked A, B, C and D. Only one of them is correct. You should decide on the correct choice and write down its mark on the answer sheet.

- SQL is _____ language.
A. a procedural B. a host C. an assemble D. a non-procedural
- Which of the following operations cannot be carried out using SQL?
A. updating views B. querying views
C. creating base tables based on views D. creating new views based on old views
- In SQL, CREATE, DROP and ALTER are _____ commands.
A. data query B. data modification C. data definition D. data control
- In SQL, in order to delete a view, we use _____ command.
A. DELETE B. DROP C. CLEAR D. REMOVE
- In SQL, views are corresponding to the _____ schema of a database.
A. external B. conceptual C. inner D. storage
- Which of the following SQL statements is not a data definition statement?
A. CREATE TABLE B. DROP VIEW
C. CREATE VIEW D. GRANT
- In SQL, in order to delete an existing base table S, we use _____ command.
A. DELETE TABLE S B. DELETE S
C. DROP TABLE S D. DROP S
- In SQL, in order to add a new column CN into base table S, we use _____ command.
A. ADD TABLE S(CN CHAR(8)) B. ADD TABLE S ALTER(CN CHAR(8))
C. ALTER TABLE S ADD CN CHAR(8) D. ALTER TABLE S (ADD CN CHAR(8))
- Given table S(Sno, Sname, Sgender, Sage), in order to delete column Sage of S using SQL, we shall use _____ command.
A. DELETE Sage FROM S B. ALTER TABLE S DROP Sage
C. UPDATE S Sage D. ALTER TABLE S 'Sage'
- Suppose we are given relation $S(Sno \text{ CHAR}(5), Sname \text{ CHAR}(10), Sage \text{ INT})$, $SC(Sno \text{ CHAR}(5), Cno \text{ CHAR}(5), Grade \text{ FLOAT})$ and $C(Cno \text{ CHAR}(5), Cname \text{ CHAR}(40))$, where a row in S means that a student with student number *Sno* and name *Sname* has age *Sage*, a row in SC means that a student with student number *Sno* takes part in a course with course number *Cno* and obtains a grade *Grade*, and a row in C means that a course with course number *Cno* has course name *Cname*, then in order to list student name of each student that takes part in course 'ACCESS' and has a age not less than 20, the possible SQL statement we may use will be SELECT SNAME FROM S, C, SC WHERE _____.
A. $S.Sno=SC.Sno \text{ AND } C.Cno=SC.Cno \text{ AND } Sage \geq 20 \text{ AND } Cname = 'ACCESS'$
B. $S.Sno=SC.Sno \text{ AND } C.Cno=SC.Cno \text{ AND } Sage \geq 20 \text{ AND } Cname \text{ IN } 'ACCESS'$
C. $Sage \geq 20 \text{ AND } Cname \text{ IN } 'ACCESS'$
D. $Sage \geq 20 \text{ AND } Cname = 'ACCESS'$
- Suppose we are given relation $S(SN \text{ CHAR}(10), CN \text{ CHAR}(40), GRADE \text{ FLOAT})$, where a row in S means that a student with student name *SN* takes part in a course with course name

- CN and obtains a grade *GRADE*, then in order to insert information “Student ‘Zhangming’ takes part in course ‘database’ and obtains a grade 80” into *S*, the possible SQL statement we may use will be ____.
- ADD INTO *S* VALUES(‘Zhangming’, ‘database’, ‘80’)
 - INSERT INTO *S* VALUES(‘Zhangming’, ‘database’, ‘80’)
 - ADD INTO *S* VALUES(‘Zhangming’, ‘database’, 80)
 - INSERT INTO *S* VALUES(‘Zhangming’, ‘database’, 80)
12. Suppose we are given relation *S*(*SN* CHAR(10), *CN* CHAR(40), *GRADE* FLOAT), where a row in *S* means that a student with student name *SN* takes part in a course with course name *CN* and obtains a grade *GRADE*, then in order to set Zhangming’s database grade to 85, the possible SQL statement we may use will be ____.
- UPDATE *S* SET *GRADE*=85 WHERE *SN*=‘Zhangming’ and *CN*=‘database’
 - UPDATE *S* SET *GRADE*=‘85’ WHERE *SN*=‘Zhangming’ and *CN*=‘database’
 - UPDATE *GRADE*=85 WHERE *SN*=‘Zhangming’ and *CN*=‘database’
 - UPDATE *GRADE*=‘85’ WHERE *SN*=‘Zhangming’ and *CN*=‘database’
13. In SQL, subquery is a query statement that ____.
- returns a subset of data in one table
 - returns a subset of fields in multiple tables
 - selects a subset of fields in one table
 - is embedded into another query statement
14. Suppose we are given relation *S*(*Sno* CHAR(5), *Sname* CHAR(10), *Sgender* INT), *SC*(*Sno* CHAR(5), *Cno* CHAR(5), *Grade* FLOAT) and *C* (*Cno* CHAR(5), *Cname* CHAR(40)), where a row in *S* means that a student with student number *Sno* and name *Sname* has sex *Sssex*, a row in *SC* means that a student with student number *Sno* takes part in a course with course number *Cno* and obtains a grade *Grade*, and a row in *C* means that a course with course number *Cno* has course name *Cname*, then in order to list student name of each male student that takes part in course ‘database’, the possible SQL statement we may use will be SELECT *SNAME* FROM *S*, *C*, *SC* WHERE ____.
- S.Sno*=*SC.Sno* and *C.Cno*=*SC.Cno* and *Sgender*=‘M’ and *Cname* = ‘database’
 - S.Sno*=*SC.Sno* and *C.Cno*=*SC.Cno* and *Sgender* in ‘M’ and *Cname* in ‘database’
 - Sgender* ‘M’ and *Cname* ‘database’
 - Sgender*=‘M’ and *Cname* = ‘database’
15. If we create a table *SC* using the following SQL statement:
- ```
CREATE TABLE SC(Sno CHAR(6) NOT NULL,
 Cno CHAR(3) NOT NULL,
 Grade INT,
 NOTE CHAR(20)),
```
- then SQL statement INSERT INTO *SC* VALUES \_\_\_\_ can be executed successfully.
- (‘201009’, ‘111’, 60, compulsory)
  - (‘200823’, ‘101’, NULL, NULL)
  - (NULL, ‘103’, 80, ‘compulsory ‘)
  - (‘201132’, NULL, 86, “)
16. Suppose we are given relation *S*(*Sno* CHAR(5), *Sname* CHAR(10), *Ssex* INT ), *SC*( *Sno* CHAR(5), *Cno* CHAR(5), *Grade* FLOAT) and *C* ( *Cno* CHAR(5), *Cname* CHAR(40) ), where a row in *S* means that a student with student number *Sno* and name *Sname* has sex *Sssex*, a row in *SC* means that a student with student number *Sno* takes part in a course with course number *Cno* and obtains a grade *Grade*, and a row in *C* means that a course with course number *Cno* has course name *Cname*, then in order to list student name of each male student

- that takes part in course 'database', tables must be involved are \_\_\_\_.
- A. S, C                      B. S, SC                      C. C, SC                      D. S, C, SC
17. In SQL, SELECT clause can realize \_\_\_\_ operation of relational algebra.  
A. projection                      B. selection                      C. join                      D. intersection
18. In SQL, WHERE clause can realize \_\_\_\_ operation of relational algebra.  
A. projection                      B. selection                      C. join                      D. intersection
19. In SQL, in order to eliminate repeated rows in query results, which of the following key word should be used?  
A. UNIQUE                      B. COUNT                      C. DISTINCT                      D. UNION
20. Which of the following SQL operation is not correct (where AGE is a field name of a table)?  
A. AGE IS NULL                      B. AGE IS NOT NULL  
C. AGE=NULL                      D. NOT ( AGE IS NULL )
21. Suppose we are given relation DEPT(DNO CHAR(5), DNAME CHAR(20) ), in order to find the DNAMEs that contain at least 4 characters and if count backwards, the 3<sup>rd</sup> alphabet of which is W, the WHERE clause may be WHERE DNAME LIKE \_\_\_\_.  
A. '\_ \_W \_%'                      B. '\_ W \_ %'                      C. '\_ W \_ \_'                      D. '\_ %W \_ \_'
22. If there are multiple tables involved in a SELECT statement, then the WHERE clause of the statement must include \_\_\_\_.  
A. only selection conditions                      B. joining conditions or selection conditions  
C. only joining conditions                      D. joining conditions and selection conditions
23. In SQL, the operation that is equivalent to NOT IN is \_\_\_\_\_.  
A. =ANY                      B. <>ANY                      C. =ALL                      D. <>ALL
24. In SQL, the operation that is equivalent to SALARY IN (1000, 2000) is \_\_\_\_\_.  
A. SALARY <=2000 AND SALARY >=1000                      B. SALARY <2000 AND SALARY >1000  
C. SALARY =2000 AND SALARY=1000                      D. SALARY =2000 OR SALARY=1000

## PART B

Fill in blanks with proper words or phrases.

- The full name of SQL is (a). The functions of SQL include: (b), (c), and (d).
- In SQL, statements that realize data query are (a) statements; (b) have corresponding physical storage, while (c) have not corresponding physical storage.
- [☆]Views are tables that are derived from (a) or (b).
- Suppose relation R(A, B, C) and S(A, D, E, F) have the same attribute A, then algebra relation expression  $\Pi_{R.A, B, D, F}(R \bowtie S)$  can be expressed using the following SQL statement:  
SELECT R.A, B, D, F  
FROM R, S  
WHERE (a).
- Suppose we are given relation R(BH, XM, XB, DWH), S(DWH, DWM) and T(BH, XM, XB, DWH), then :  
(a) the SQL statement that realize  $R \cup T$  is (a);  
(b) the SQL statement that realize  $\sigma_{DWH=100}(R)$  is (b);  
(c) the SQL statement that realize  $\Pi_{XM, XB}(R)$  is (c);  
(d) the SQL statement that realize  $\Pi_{XM, DWH}(\sigma_{XB='F'}(R))$  is (d);  
(e) the SQL statement that realize  $R \bowtie S$  is (e);

- (f) the SQL statement that realize  $\Pi_{XM, XB, DWH}(\sigma_{XB='M'}(R \bowtie S))$  is (f).
6. Suppose we are given student relation S(No, Name, Gender, Age, Class), in which No is the student number, Name is the student name, Sex is the student sex, Age is the student age, Class is the class number, No and Age are integers, others are characters, then :
- (a) the SQL statement that inserts record (25, 'Liming', 'M', 21, '95031') is (a).
- (b) the SQL statement that inserts a student with student number 30, name 'Zhenghe' and class number '95031' is (b).
- (c) for a student with student number 10, the SQL statement that modifies his or her name to 'Wanghua' and modifies his or her gender to 'F' is (c).
- (d) the SQL statement that modifies class number '95101' to '95091' is (d).
- (e) the SQL statement that deletes the record of a student with student number 20 is (e).
- (f) the SQL statement that deletes all the records of students with family name 'Wang' is (f).
7. The manager of a car showroom uses a database to store data about cars he sells. This is part of the database.

| Maker      | Model  | Size of engine | Registration | Price |
|------------|--------|----------------|--------------|-------|
| Opel       | Vectra | 1.8            | VSE 648      | 19000 |
| Opel       | Zafira | 2.0            | BFK 297      | 29000 |
| Volkswagen | Golf   | 1.4            | SB A5526     | 15000 |
| Volkswagen | Polo   | 1.2            | DD B4978     | 11000 |
| Volkswagen | Jetta  | 1.6            | B G8347      | 19000 |
| Renault    | Megane | 1.4            | 1233 CD 33   | 17000 |
| Renault    | Clio   | 1.2            | 6289 XF 54   | 11000 |

- (1) There are (a) records in this part of the database.
- (2) There are (b) fields in this part of the database.
- (3) If we sort the records in descending order of their Size of engine, then the Registration of the first record in the database after it has been sorted is (c).

## PART C

### Answer the following questions.

1. What is a view? Discuss the difference between a view and a base relation.